10.3.1 Fitting of Binomial Distribution

When a Binomial distribution is to be fitted to an observed data the following procedure is adopted:-

(i) Find mean $\bar{x} = \frac{\sum fx}{\sum f} = np$

(ii) Find $p = \frac{\bar{x}}{n}$

(iii) Find $q = 1 - p$

(iv) Write the probability mass function : $P(x) = nC_x p^x q^{n-x}$ $x = 0, 1, 2, \ldots n$

(v) Put $x = 0$; find $P(0) = nC_0 p^0 q^{n-0} = q^n$

(vi) Find the expected frequency of $X = 0$ i.e., $F(0) = N \times P(0)$, where $N = \sum f_i$

(vii) The other expected frequencies are obtained by using the recurrence formula

$$F(x+1) = \frac{n-x}{x+1} \times \frac{p}{q} \times F(x)$$

Example 10.34

A set of three similar coins are tossed 100 times with the following results

Number of heads	0	1	2	3
Frequency	36	40	22	2

Fit a binomial distribution and estimate the expected frequencies.

Solution :

x	f	fx
0	36	0
1	40	40
2	22	44
3	2	44
Total	100	90

(i) Mean $\bar{x} = \frac{\sum fx}{\sum f} = \frac{90}{100} = 0.9$

(ii) $p = \frac{\bar{x}}{n} = \frac{0.9}{3} = 0.3$

(iii) $q = 1 - p = 1 - 0.3 = 0.7$

(iv) $P(x) = nC_x p^x q^{n-x} = 3C_x\ 0.3_x\ 0.7^{3-x}$